

Artificial Intelligence and Robotics

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FOREWORD

Welcome to the UK-RAS White Paper Series on Robotics and Autonomous Systems (RAS). This is one of the core activities of UK-RAS Network, funded by the Engineering and Physical Sciences Research Council (EPSRC). By bringing together academic centres of excellence, industry, government, funding bodies and charities, the Network provides academic leadership, expands collaboration with industry while integrating and coordinating activities at EPSRC funded RAS capital facilities, Centres for Doctoral Training and partner universities.

The recent successes of AI have captured the wildest imagination of both the scientific communities and the general public. Robotics and AI amplify human potentials, increase productivity and are moving from simple reasoning towards human-like

cognitive abilities. Current AI technologies are used in a set area of applications, ranging from healthcare, manufacturing, transport, energy, to financial services, banking, advertising, management consulting and government agencies. The global AI market is around 260 billion USD in 2016 and it is estimated to exceed 3 trillion by 2024. To understand the impact of AI, it is important to draw lessons from its past successes and failures and this white paper provides a comprehensive explanation of the evolution of AI, its current status and future directions.

The UK-RAS white papers are intended to serve as a basis for discussing the future technological roadmaps, engaging the wider community and stakeholders, as well as policy makers, in assessing the potential social, economic and ethical/legal impact

of RAS. It is our plan to provide annual updates for these white papers so your feedback is essential - whether it is to point out inadvertent omissions of specific areas of development that need to be covered, or to suggest major future trends that deserve further debate and in-depth analysis.

Please direct all your feedback to white-paper@ukras.org. We look forward to hearing from you!



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On behalf of the UK-RAS Network, established to provide academic leadership, expand collaboration with industry while integrating and coordinating activities at EPSRC funded RAS capital facilities, Centres for Doctoral Training and partner universities.

1. INTRODUCTION

Artificial Intelligence (AI) is a commonly employed appellation to refer to the field of science aimed at providing machines with the capacity of performing functions such as logic, reasoning, planning, learning, and perception. Despite the reference to “machines” in this definition, the latter could be applied to “any type of living intelligence”. Likewise, the meaning of intelligence, as it is found in primates and other exceptional animals for example, it can be extended to include an interleaved set of capacities, including creativity, emotional knowledge, and self-awareness.

The term AI was closely associated with the field of “symbolic AI”, which was popular until the end of the 1980s. In order to overcome some of the limitations of symbolic AI, subsymbolic methodologies such as neural networks, fuzzy systems, evolutionary computation and other computational models started gaining popularity, leading to the term “computational intelligence” emerging as a subfield of AI.

Nowadays, the term AI encompasses the whole conceptualisation of a machine that is intelligent in terms of both operational and social consequences. A practical definition used is one proposed by Russell and Norvig: “Artificial Intelligence is the study of human intelligence and actions replicated artificially, such that the resultant bears to its design a reasonable level of rationality” [1]. This definition can be further refined by stipulating that the level of rationality may even supersede humans, for specific and well-defined tasks.

Current AI technologies are used in online advertising, driving, aviation, medicine and personal assistance image recognition. The recent success of AI has captured the imagination of both the scientific community and the public. An example of this is vehicles equipped with an automatic steering system, also known as autonomous cars. Each vehicle is equipped with a series of lidar sensors and cameras which enable recognition of its three-dimensional environment and provides the ability to make intelligent

decisions on maneuvers in variable, real-traffic road conditions. Another example is the Alpha-Go, developed by Google Deepmind, to play the board game Go. Last year, Alpha-Go defeated the Korean grandmaster Lee Sedol, becoming the first machine to beat a professional player and recently it went on to win against the current world number one, Ke Jie, in China. The number of possible games in Go is estimated to be 10^{761} and given the extreme complexity of the game, most AI researchers believed it would be years before this could happen. This has led to both the excitement and fear in many that AI will surpass humans in all the fields it marches into.

However, current AI technologies are limited to very specific applications. One limitation of AI, for example, is the lack of “common sense”; the ability to judge information beyond its acquired knowledge. A recent example is that of the AI robot Tay developed by Microsoft and designed for making conversations on social networks. It had to be disconnected shortly after its launch because it was not able to distinguish between positive and negative human interaction. AI is also limited in terms of emotional intelligence. AI can only detect basic human emotional states such as anger, joy, sadness, fear, pain, stress and neutrality. Emotional intelligence is one of the next frontiers of higher levels of personalisation.

True and complete AI does not yet exist. At this level, AI will mimic human cognition to a point that it will enable the ability to dream, think, feel emotions and have own goals. Although there is no evidence yet this kind of true AI could exist before 2050, nevertheless the computer science principles driving AI forward, are rapidly advancing and it is important to assess its impact, not only from a technological standpoint, but also from a social, ethical and legal perspective.

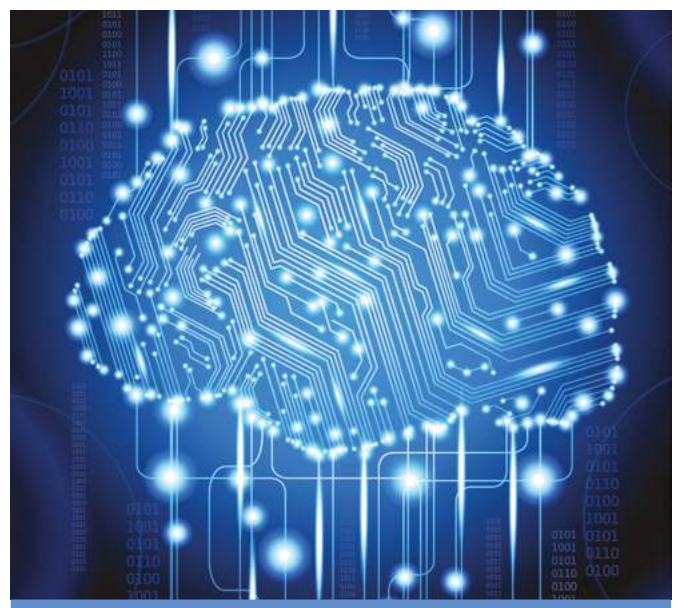
In 1971, DARPA funded a consortium of leading laboratories in the field of speech recognition. The project had the ambitious goal of creating a fully functional speech recognition system with a large vocabulary. In the middle of the 1970s, the field of AI endured fierce criticism and budgetary restrictions, as AI research development did not match the overwhelming expectations of researchers.. When promised results did not materialize, investment in AI eroded. Following disappointing results, DARPA withdrew funding in speech recognition and this, coupled with other events such as the failure of machine translation, the abandonment of connectionism and the Lighthill report, marked the first winter of AI [2]. During this period, connectionism stagnated for the next 10 years following a devastating critique by Marvin Minsky on perceptrons [3].

From 1980 until 1987, AI programmes, called “expert systems”, were adopted by companies and knowledge acquisition became the central focus of AI research. At the same time, the Japanese government launched a massive funding program on AI, with its fifth-generation computers initiative. Connectionism was also revived by the work of John Hopfield [4] and David Rumelhart [5].

AI researchers who had experienced the first backlash in 1974, were sceptical about the reignited enthusiasms of expert systems and sadly their fears were well founded. The first sign of a changing tide was with the collapse of the AI computer hardware market in 1987. Apple and IBM desktops had gradually improved their speed and power and in 1987 they were more powerful than the best LISP machines on the market. Overnight however, the industry

collapsed and billions of dollars were lost. The difficulty of updating and reprogramming the expert systems, in addition to the high maintenance costs, led to the second AI winter. Investment in AI dropped and DARPA stopped its strategic computing initiative, claiming AI was no longer the “latest mode”. Japan also stopped funding its fifth-generation computer program as the proposed goals were not achieved.

In the 1990s, the new concept of “intelligent agent” emerged [6]. An agent is a system that perceives its environment and undertakes actions that maximize its chances of being successful. The concept of agents conveys, for the first time, the idea of intelligent units working collaboratively with a common objective. This new paradigm was intended to mimic how humans work collectively in groups, organizations and/or societies. Intelligent agents proved to be a more polyvalent concept of intelligence. In the late 1990s, fields such as statistical learning from several perspectives including probabilistic, frequentist and possibilistic (fuzzy logic) approaches, were linked to AI to deal with the uncertainty of decisions. This brought a new wave of successful applications for AI, beyond what expert systems had achieved during the 1980s. These new ways



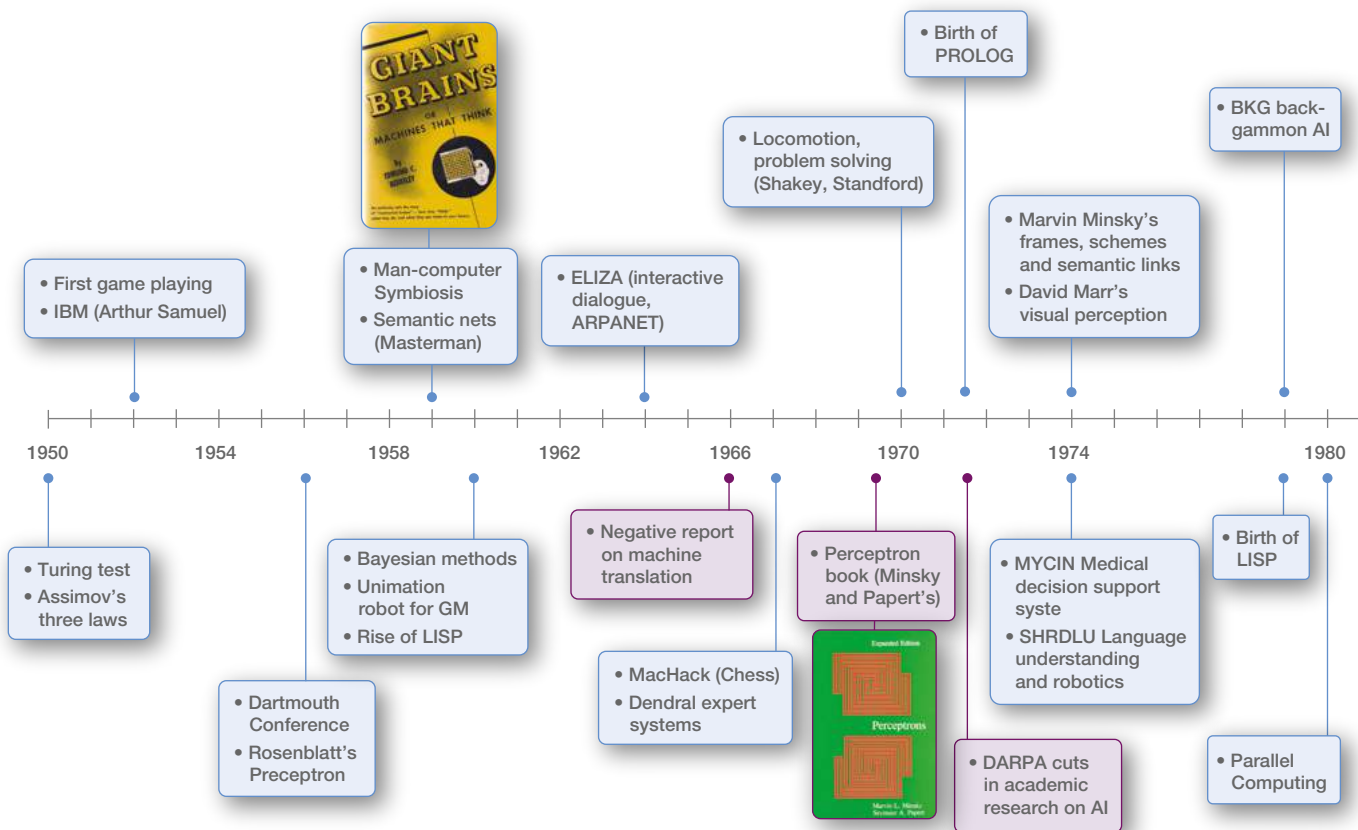
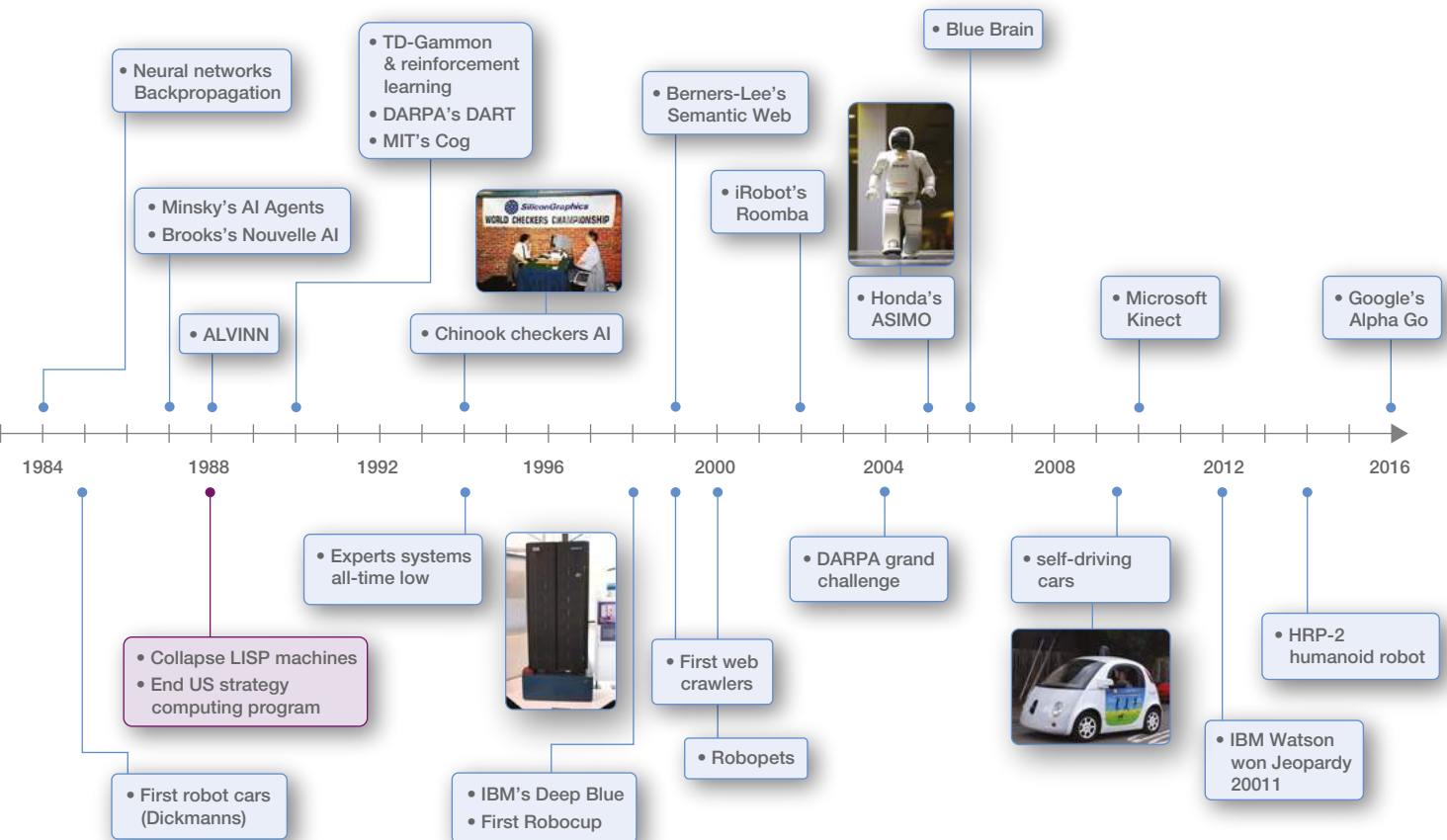


Figure 1.

A timeline highlighting some of the most relevant events of AI since 1950. The blue boxes represent events that have had a positive impact on the development of AI. In contrast, those with a negative impact are shown in red and reflect the low points in the evolution of the field, i.e. the so-called "winters" of AI.

of reasoning were more suited to cope with the uncertainty of intelligent agent states and perceptions and had its major impact in the field of control. During this time, high-speed trains controlled by fuzzy logic, were developed [7] as were many other industrial applications (e.g. factory valves, gas and petrol tanks surveillance, automatic gear transmission systems and reactor control in power plants) as well as household appliances with advanced levels of intelligence (e.g. air-conditioners, heating systems, cookers and vacuum-cleaners). These were different to the expert systems in 1980s; the modelling of the inference system for the task, achieved through learning, gave rise to the field of Machine Learning. Nevertheless, although machine reasoning exhibited good performance, there was still an engineering requirement to digest the input space into a new source, so

that intelligence could reason more effectively. Since 2000, a third renaissance of the connectionism paradigm arrived with the dawn of Big Data, propelled by the rapid adoption of the Internet and mobile communication. Neural networks were once more considered, particularly in the role they played in enhancing perceptual intelligence and eliminating the necessity of feature engineering. Great advances were also made in computer vision, improving visual perception, increasing the capabilities of intelligent agents and robots in performing more complex tasks, combined with visual pattern recognition. All these paved the way to new AI challenges such as, speech recognition, natural language processing, and self-driving cars. A timeline of key highlights in the history of AI is shown in Figure 1.



Weak and Strong AI

When defining the capacity of AI, this is frequently categorised in terms of weak or strong AI. Weak AI (narrow AI) is one intended to reproduce an observed behaviour as accurately as possible. It can carry out a task for which they have been precision-trained. Such AI systems can become extremely efficient in their own field but lack generalisational ability. Most existing intelligent systems that use machine learning, pattern recognition, data mining or natural language processing are examples of weak AI. Intelligent systems, powered with weak AI include recommender systems, spam filters, self-driving cars, and industrial robots. Strong AI is usually described as an intelligent system endowed with real consciousness and is able to think

and reason in the same way as a human being. A strong AI can, not only assimilate information like a weak AI, but also modify its own functioning, i.e. is able to autonomously reprogram the AI to perform general intelligent tasks. These processes are regulated by human-like cognitive abilities including consciousness, sentience, sapience and self-awareness. Efforts intending to generate a strong AI have focused on whole brain simulations, however this approach has received criticism, as intelligence cannot be simply explained as a biological process emanating from a single organ but is a complex coalescence of effects and interactions between the intelligent being and its environment, encompassing a series of diverse ways via interlinked biological process.

3. QUESTIONING THE IMPACT OF AI

Given the exponential rise of interest in AI, experts have called for major studies on the impact of AI on our society, not only in technological but also in legal, ethical and socio-economic areas. This response also includes the speculation that autonomous super artificial intelligence may one day supersede the cognitive capabilities of humans. This future scenario is usually known in AI forums as the “AI singularity” [8]. This is commonly defined as the ability of machines to build better machines by themselves. This futuristic scenario has been questioned and is received with scepticism by many experts. Today’s AI researchers are more focused on developing systems that are very good at tasks in a narrow range of applications. This focus is at odds with the idea of the pursuit of a super generic AI system that could mimic all different cognitive abilities related to human intelligence such as self-awareness and emotional knowledge. In addition to this debate, about AI development and the status of our hegemony as the most intelligent species on the planet, further societal concerns have been raised. For example, the AI100 (One Hundred Year Study on Artificial Intelligence) a committee led by Stanford University, defined 18 topics of importance for AI [9]. Although these are not exhaustive nor definitive, it sets forth the range of topics that need to be studied, for the potential impact of AI and stresses that there are a number of concerns to be addressed. Many similar assessments have been performed and they each outline similar concerns related to the wider adoption of AI technology.



The 18 topics covered by the AI100

Technical trends and surprises: This topic aims at forecasting the future advances and competencies of AI technologies in the near future. Observatories of the trend and impact of AI should be created, helping to plan the setting of AI in specific sectors, and preparing the necessary regulation to smooth its introduction.

Key opportunities for AI: How advances in AI can help to transform the quality of societal services such as health, education, management and government, covering not just the economic benefits but also the social advantages and impact.

Delays with translating AI advances into real-world values: The pace of translating AI into real world applications is currently driven by potential economic prospects [10]. It is necessary to take measures to foster a rapid translation of those potential applications of AI that can improve or solve a critical need of our society, such as those that can save lives or greatly improve the organisation of social services, even though their economic exploitation is not yet assured.

Privacy and machine intelligence: Personal data and privacy is a major issue to consider and it is important to envisage and prepare the regulatory, legal and policy frameworks related to the sharing of personal data in developing AI systems.

Democracy and freedom: In addition to privacy, ethical questions with respect to the stealth use of AI for unscrupulous applications must be considered. The use of AI should not be at the expense of limiting or influencing the democracy and the freedom of people.

Law: This considers the implications of relevant laws and regulations. First, to identify which aspects of AI require legal assessment and what actions should be undertaken to ensure law enforcement for AI services. It should also provide frameworks and guidelines about how to adhere to the approved laws and policies.

Ethics: By the time AI is deployed into real world applications there are ethical concerns referring to their interaction with the world. What uses of AI should be considered unethical? How should this be disclosed?

It is also important to note that the funding in AI safety, ethics and strategy/policy has almost doubled in the last three years [24]. Apart from non-profit organisations, such as the Future of Life Institute (FLI) and the Machine Intelligence Research Institute (MIRI), other centres, such as the Centre for Human-Compatible AI and Centre for the Future of Intelligence, have emerged and they, along with key technological firms, invested a total of \$6.6 million in 2016.

4.3 PUBLICATION VERSUS PATENTING

In terms of international output in publications and patents, there has been a shift of predominant countries influencing the field of AI. In 1995 USA and Japan were the two leading countries in the field of AI patents but this has now shifted to

Asia, with China becoming a major player. Since 2010 China and USA have led the way in both scientific publications and in the filing of patents. Other emerging economies, such as India and Brazil, are also rapidly rising.

Recently, there has been a shift relocation of many academic professionals in AI to the industrial sector. The world's largest technology companies have hired several entire research teams previously in universities. These corporations have opted for the publication of pre-prints (ArXiv¹ or viXra²) and other non-citable documents, instead of using conventional academic methods of peer-review. This has become an increasing trend. The reason for this is that it allows the prioritisation of claim imprinting without the delay of a peer-review process.

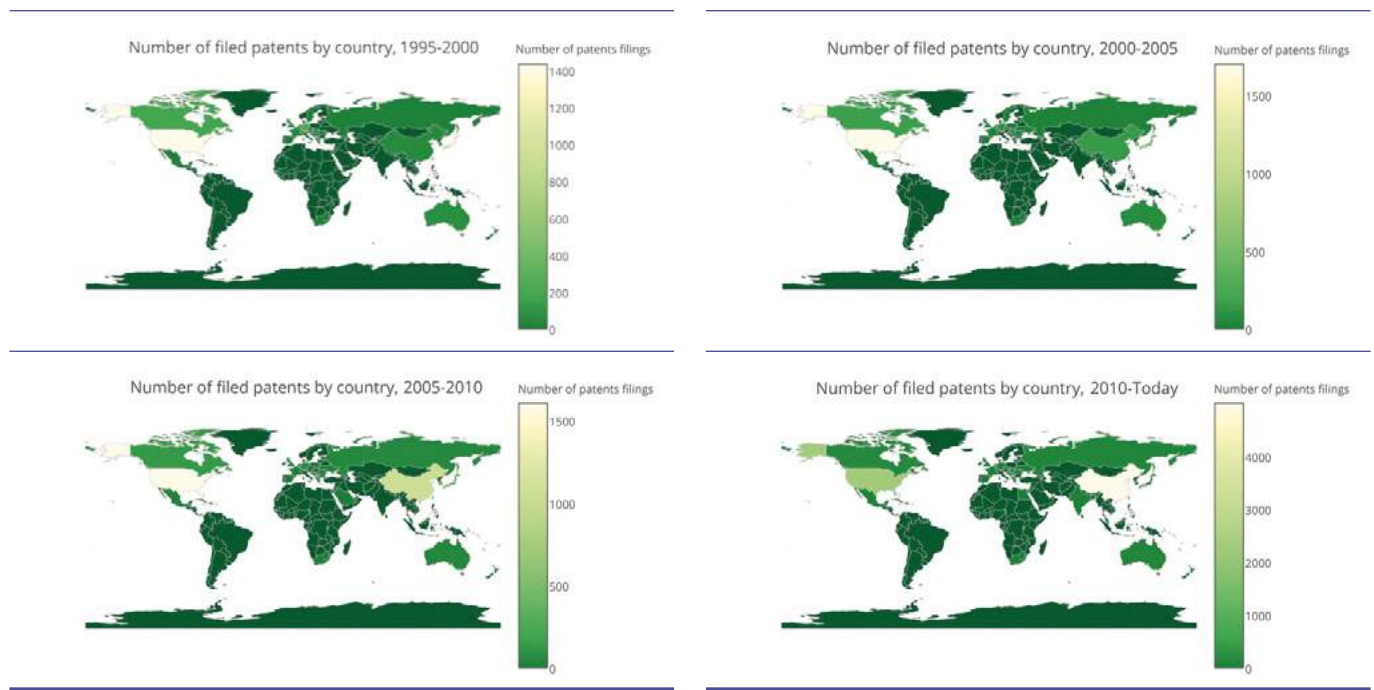


Figure 2
Evolution of filed patents in AI across countries for five-year periods since 1995 to present day.

¹ <https://arxiv.org/>

² vixra.org

offers Cortana, and Google has the Google Assistant. In 2016, Apple Inc. purchased Emotient Inc., a start-up using artificial-intelligence technology to read people's emotions by analyzing facial expressions. DeepMind created WaveNet, which is a generative model that mimics human voices. According to the company's website, this sounds more natural than the best existing Text-to-Speech systems. Facebook is also considering machine-human interaction capabilities as a prerequisite to generalised AI.

Recently, OpenAI, a non-profit organisation, has been funded as part of a strategic plan to mitigate the risks of monopolising strong AI. OpenAI has re-designed evolutionary algorithms that can work together with deep neural networks to offer state-of-the-art performance. It is considered to rival DeepMind since it offers similar open-source machine learning libraries to TensorFlow, a deep learning library distributed by Google DeepMind. Nevertheless, the big difference between the technology developed at OpenAI and the other private tech companies, is that the created Intellectual Property is accessible by everyone.

Although several companies and organisations, including DeepMind and OpenAI, envision the solution to the creation of intelligence and the so-called Strong AI, developing machines with self-sustained long-term goals is well beyond current technology. Furthermore, there is vigorous debate on whether or not we are going through an AI bubble, which encompasses the paradox that productivity growth in USA, during the last decade, has declined regardless of an explosion of technological progress and innovation. It is difficult to understand whether this reflects a statistical shortcoming or that current innovations are not transformative enough. This decline can be also attributed to the lack of consistent policy frameworks and security standards that can enable the application of AI in projects of significant impact.



7. THE RISE OF DEEP LEARNING: RETHINKING THE MACHINE LEARNING PIPELINE

The idea of creating an artificial machine is as old as the invention of the computer. Alan Turing in the early 1950s proposed the Turing test, designed to assess whether a machine could be defined as intelligent. Two of the main pioneers in this field are Pitts and McCulloch [38] who, in 1943, developed a technique designed to mimic the way a neuron works. Inspired by this work, a few years later, Frank Rosenblatt [39] developed the first real precursor of the modern neural network, called Perceptron. This algorithm describes an automatic learning procedure that can discriminate linearly separable data. Rosenblatt was confident that the perceptron would lead to an AI system in the future. The introduction of perceptron, in 1958, signalled the beginning of the AI evolution. For almost 10 years afterwards, researchers used this approach to automatically learn how to discriminate data in many applications, until Papert and Minsky [3], demonstrated a few important limitations of Perceptron. This slowed down the fervour of AI progress and more specifically, they proved that the perceptron was not capable of learning simple functions, such as the exclusive-or XOR, no matter how long the network was trained.

Today, we know that the model implied by the perceptron is linear and the XOR function does not belong to this family, but at the time this was enough to stop the research behind neural nets and began the first AI winter. Much later in 1974, the idea of organizing the perceptron in layers and training them using the delta rule [40] shaped the creation of more complex neural nets. With the introduction of the Multilayer Neural Nets [41], researchers were confident that adding multiple hidden layers to the networks would produce deep architectures that further increase the complexity of the hypothesis that can be expressed. However, the hardware constraints that were present at that time, limited, for many years, the number of layers that could be used in practice. To overcome these hardware limitations different network configurations were proposed. For almost another decade, researchers focused on producing new efficient network architectures that are suitable for specific contexts. Notably, these developments included the Autoencoder [42] useful in extracting relevant features from data, the Belief nets used to model statistical variables, the Recurrent neural nets [43] and its variant Long Short Term Memory [44] used for processing sequence of data, and the Convolutional neural nets [45] used to process images. Despite these new AI solutions, the aforementioned hardware limitations were a big restriction during training.

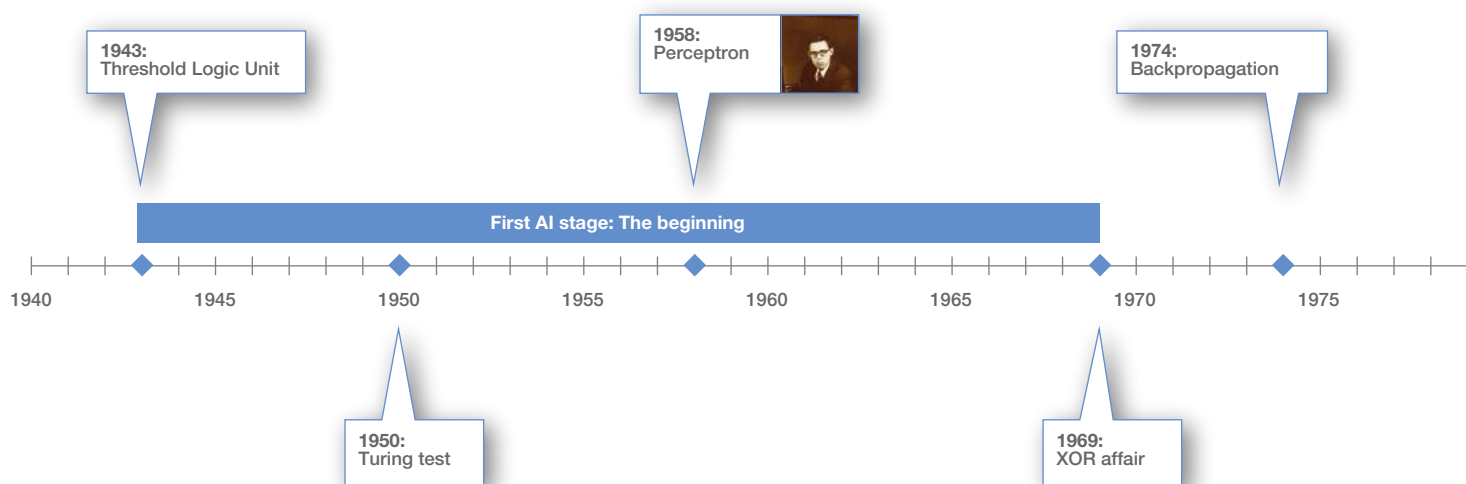


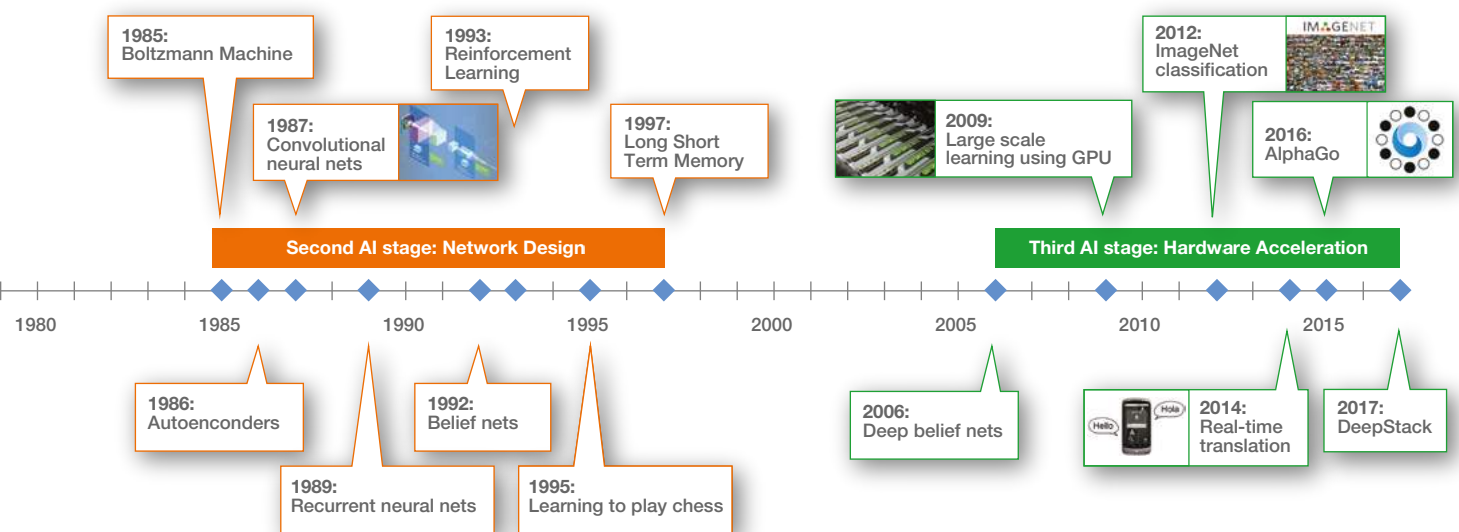
Figure 6

Timeline of the main discoveries and seasons in AI: from perceptron to deep learning

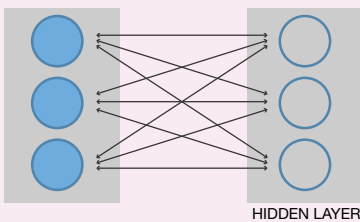
With recent hardware advances, such as the parallelization using GPU, the cloud computing and the multi-core processing finally led to the present stage of AI. In this stage, deep neural nets have made tremendous progress in terms of accuracy and they can now recognize complex images and perform voice translation in real time. However, researchers are still dealing with issues relating to the overfitting of the networks, since large datasets are often required and not always available. Furthermore, with the vanishing of the gradient, this leads to a widespread problem generated during the training of a network with many layers. For this reason, more sophisticated training procedures have recently been proposed. For example, in 2006, Hinton introduced the idea of unsupervised pretraining and Deep Belief Nets [46]. This approach has each pair of consecutive layers trained separately using an unsupervised model similar to the one used in the Restricted Boltzman Machine [47]; then the obtained parameters are frozen and a new pair of layers are trained and stacked on top of the previous ones. This procedure can be repeated many times leading to the development of a deeper architecture with respect to the traditional neural nets. Moreover, this unsupervised pre-training approach has led to increasing neural net papers when in 2014, for the first time, a neural model became state-of-the-art in the speech recognition.

In 2010, a large database, known as Imagenet containing millions of labelled images was created and this was coupled with an annual challenge called Large Scale Visual Recognition Challenge. This competition requires teams of researchers to build AI systems and they receive a score based on how accurate their model is. In the first two years of the contest, the top models had an error rate of 28% and 26%. In 2012, Krizhevsky, Sutskever and Hinton [48] submitted a solution that had an error rate of just 16% and in 2015 the latest submitted models [49] were capable of beating the human experts with an overall error of 5%. One of the main components of this significant improvement, in such a short time, was due to the extensive use of graphics processing units (GPUs) for speeding up the training procedure, thus allowing the use of larger models which also meant a lower error rate in classification.

In the last 3 years researchers have also been working on training deep neural nets that are capable of beating human experts in different fields, similar to the solution used for AlphaGo [50] or DeepStack [51] and in 2017, they overtook human experts with 44000 played hands of poker.

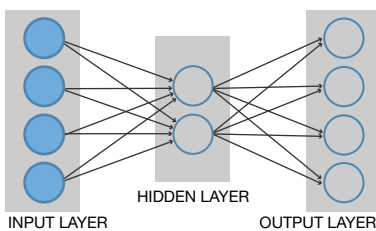


**1985:
Boltzmann Machines**



Boltzmann Machines represent a type of neural network modelled by using stochastic units with a specific distribution (for example Gaussian). Learning procedure involves several steps called Gibbs sampling, which gradually adjust the weights to minimize the reconstruction error. They are useful if it is required to model probabilistic relationships between variables. A variant of this machine is the Restricted Boltzmann Machines where the visible and hidden units are restricted to form a bipartite graph that allows implementation of more efficient training algorithms.

**1986:
AutoEncoder**



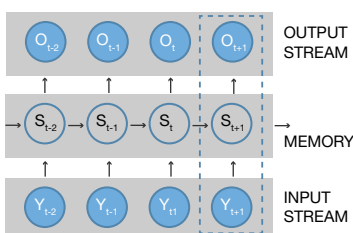
An Autoencoder is a neural network designed to extract features directly from the data. This network has the same number of input and output nodes and it is trained using an unsupervised approach to recreate the input vector rather than to assign a class label to it. Usually, the number of hidden units is smaller than the input/output layers, which achieve encoding of the data in a lower dimensional space and extract the most discriminative features.

**1987:
Convolutional neural network (CNN)**



CNNs have been proposed to process efficiently imagery data. The name of these networks comes from the convolution operator that provides an easy way to perform complex operations using convolution filter. CNNs use locally connected neurons that represent data specific kernels. The main advantage of a CNN is that during back-propagation, the network has to adjust a number of parameters equal to a single instance of the kernel which drastically reduces the connections from the typical neural network. The concept of CNN is inspired by the neurobiological model of the visual cortex and can be briefly summarized as a sequence of convolution and subsampling of the image until high level features can be extracted.

**1989:
Recurrent neural nets (RNN)**



RNN is a neural network that contains hidden units capable of analysing streams of data. Since RNN suffers from the vanishing gradient and exploding gradient problems, a variation called Long Short-Term Memory units (LSTMs) was proposed in 1997 to solve this problem. Specifically, LSTM is particularly suitable for applications where there are very long time lags of unknown sizes between important events. RNN and LSTM share the same weights across all steps that greatly reduce the total number of parameters that the network needs to learn. RNNs have shown great successes in many Natural Language Processing tasks such as language modelling, bioinformatics, speech recognition and generating image description.

Modern machines combine powerful multicore CPUs with dedicated hardware designed to solve parallel processing. GPU and FPGA are the most popular dedicated hardware commonly available in workstations developing AI systems.

A GPU (Graphics Processing Unit) is a chip designed to accelerate the processing of multidimensional data such as an image. A GPU consists of thousands of smaller cores, intended to work independently on a subspace of the input data, that needs heavy computation. Repetitive functions that can be applied to different parts of the input, such as texture mapping, image rotation, translation and filtering, are performed at a much faster rate and more efficiently, through the use of the GPU. A GPU has dedicated memory and the data must be moved in and out in order to be processed.

FPGA (Field Programmable Gate Array) is a reconfigurable digital logic containing an array of programmable logic blocks and a hierarchy of reconfigurable interconnections. An FPGA is not a processor and therefore it cannot run a program stored in the memory. An FPGA is configured using a hardware description language (HDL) and unlike the traditional CPU, it is truly parallel. This means, that each independent processing task is assigned to a dedicated section of the chip and many parts of the same program can be performed in simultaneously. A typical FPGA may also have dedicated memory blocks, digital clock manager, IO banks and several other features, which vary across different models. While a GPU is designed to perform efficiently, with similar threads on different subsets of the input, an FPGA is designed to parallelize sequential serial processing of the same program.

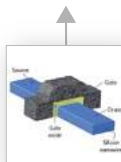
1960 Central Processing Unit (CPU):
Integrated circuits that allow large number of transistors to be manufactured in a single chip.



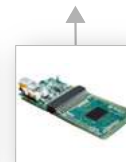
2001 Graphics Processing Unit (GPU):
Nvidia introduced the term GPU for integrated chips with image formation/rendering engines.



2010 Junction less transistor:
A control gate wrapped around a silicon nanowire that can control the passage of electrons without the use of junctions or doping.



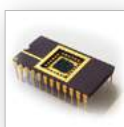
2014 TrueNorth –Neuromorphic Adaptive Plastic Scalable Electronics:
The first neuromorphic integrated circuit to achieve one million individually programmable neurons with 256 individually programmable synapses.



1990: Field Programmable Gate Array (FPGA):
Integrated circuits designed to be configured after manufacturing.



2008 Memristor:
A fourth basic passive circuit element. The memristor's properties permit the creation of smaller and better-performing electronic devices.



2011 single-electron transistor:
1.5 nanometres in diameter, made out of oxide based materials. Could spur the creation of microscopic computers.

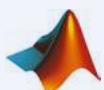


2015 3D Xpoint:
Non-volatile memory claimed to be significantly faster.



Figure 8

Timeline of the main hardware discoveries that have influenced the evolution of an AI system

Logo	Language	Date	Type	Influenced by	AI resources
	Lua		(procedural, functional)	C++, Scheme	• Versatile and lightweight language. • It is the de-facto language used for the machine and deep learning framework Torch.
	Matlab	1993	Multi-paradigm	APL	• Solid Integrated environment. Matrix, linear algebra oriented language. • A selection of toolboxes and utilities for machine learning, statistics and signal processing.
	Prolog	1984	(procedural, functional)	Planner	• Good set of utilities for expressing the relationships between objects and symbolic computation. • Large set of internal functionalities to perform logic programming.
	Python	1972	Procedural	C++, java, haskell, perl	• Highly useful standard library that makes the language versatile and flexible. Focus on rapid development. • Plethora of frameworks and utilities for AI, ML, deep learning, scientific computing, natural processing language etc.
	R	1972	Declarative	Lisp, Scheme	• Most comprehensive sets of statistical analysis functions and packages. • Rich community of tools for AI or ML provided freely through the CRAN repository.
	Scala	1993	Multiparadigm (procedural, functional)	Erlan, Haskell, Java, Lisp, Lisp (Scheme)	• Fast run time (almost as C). It runs on top of the JVM. Very good support for distributed systems. • Several libraries and frameworks for AI, ML and numerical computing (ScalaNLP).

Programming languages played a major role in the evolution of AI since the late 1950s and several teams carried out important research projects in AI; e.g. automatic demonstration programs and game programs (Chess, Ladies) [65]. During these periods researchers found that one of the special requirements for AI is the ability to easily manipulate symbols and lists of symbols rather than processing numbers or strings of characters. Since the languages of the time did not offer such facilities, a researcher from MIT, John MacCarthy, developed, during 1956-58, the definition of an ad-hoc language for logic programming, called LISP (LIST Processing language). Since then, several hundred derivative languages,

so-called "Lisp dialects", have emerged (Scheme, Common Lisp, Clojure); Indeed, writing a LISP interpreter is not a hard task for a Lisp programmer (it involves only a few thousand instructions) compared to the development of a compiler for a classical language (which requires several tens of thousands of instructions). Because of its expressiveness and flexibility, LISP was very successful in the artificial intelligence community until the 1990s.

Another important event at the beginning of AI was the creation of a language with the main purpose of expressing logic rules and axioms. Around 1972 a new language was created by Alain Colmerauer and Philippe Roussel named

13. ETHICAL AND LEGAL QUESTIONS OF AI

13.1 ETHICAL ISSUES IN ARTIFICIAL INTELLIGENCE

Threat to Privacy

Data is the “fuel” of AI and special attention needs to be paid to the information source and if privacy is breached. Protective and preventive technologies need to be developed against such threats. Although solutions to this issue may be unconnected to the AI operation per se, it is the responsibility of AI operators to make sure that data privacy is protected. Additionally, applications of AI, which may compromise the rights to privacy, should be treated with special legislation that protects the individual.

Threats to security and weaponisation of AI

With the proliferation of security risks, such as terrorism and regional conflicts, we are immersed in a global arms race that has developed a demand for new weapons powered by AI, such as autonomous drones and missiles, as well as virtual bots and malicious software aimed at sophisticated espionage. This could lead to the possibility of an unprecedented war escalation. The danger of AI is that we could potentially lose control over it. This has led associations and non-governmental organizations (NGOs) to carry out awareness-raising actions to contain the use of these military robots and potentially ban the use of such weapons.

The US military has just released a prospective document entitled ‘Robotic and Autonomous Systems Strategy’, which details its strategy for robotics and autonomous systems. The use of robotics and autonomous systems by the US military defines the following five objectives: 1) Increase knowledge abilities in operations’ theatres, 2) Reduce the amount of charge carried by the soldier; 3) Improve logistics capacity; 3) Facilitate movement and manoeuvring; 4) Increase the protection of forces. Although offensive capacity is not mentioned, the four points introduced by the US military are ambiguous with respect to their scope and limitations.

Economics and Employment Issues

Currently, 8% of jobs are occupied by robots, but in 2020 this percentage will rise to 26%. Robots will become increasingly autonomous and be able to interact, execute and make more complex decisions. Thanks to ‘big data’, robots now have a formidable database that allows them to experiment and learn which algorithms work best. The accelerated process of technological development now allows labor to be replaced by capital (machinery). However, there is a negative correlation between the probability of automation of a profession and its average annual salary, suggesting a possible increase in short-term inequality [76]. The problem is not the number of jobs lost through automation, but in creating enough to compensate for potential job losses. In previous industrial revolutions, new industries hired more people than those who lost their jobs in companies that closed, because they could not compete with the speed of development in new technologies [77]. An important note, about this revolution, is the fact that it is not only the manual trades likely to be automated, but also in jobs involving tasks of an intermediate nature, such as secretarial, administration and other office work. To deal with this situation, it is necessary to put in place legal frameworks that make sure the benefits of automation do not solely go to the employer but are distributed equally, to guarantee the maintenance of education, health and pensions.

Human Bias in Artificial Intelligence

In an article published by Science magazine, researchers saw how machine learning technology reproduces human bias, for better or for worse. Words related to the lexical domain of flowers are associated with terms related to happiness and pleasure (freedom, love, peace, joy, paradise, etc.). The words relating to insects are, conversely, close to negative terms (death, hatred, ugly, illness, pain, etc.).

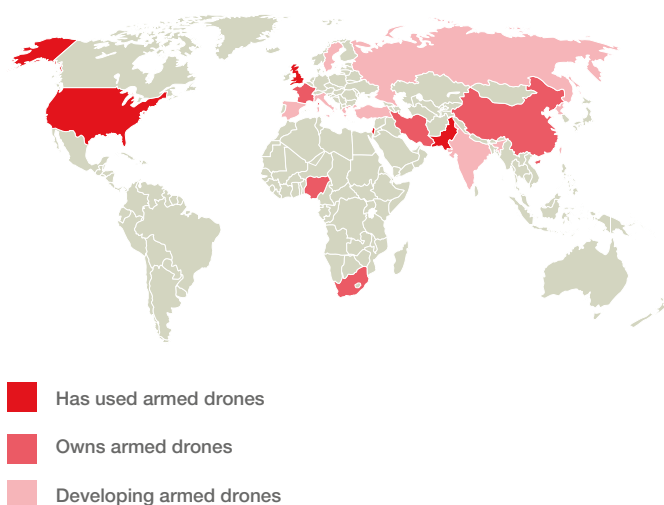


Figure 12
Countries using, owning or developing armed drones